

Project Design Phase-II
Technology Stack (Architecture & Stack)
Project Name: India's Agricultural Crop Production Analysis

Date	31 January 2026
Team ID	IACPA-01
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table-1 & table 2

Example- Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/build-a-smart-iot-enabled-system-for-order-processing-during-pandemics/>

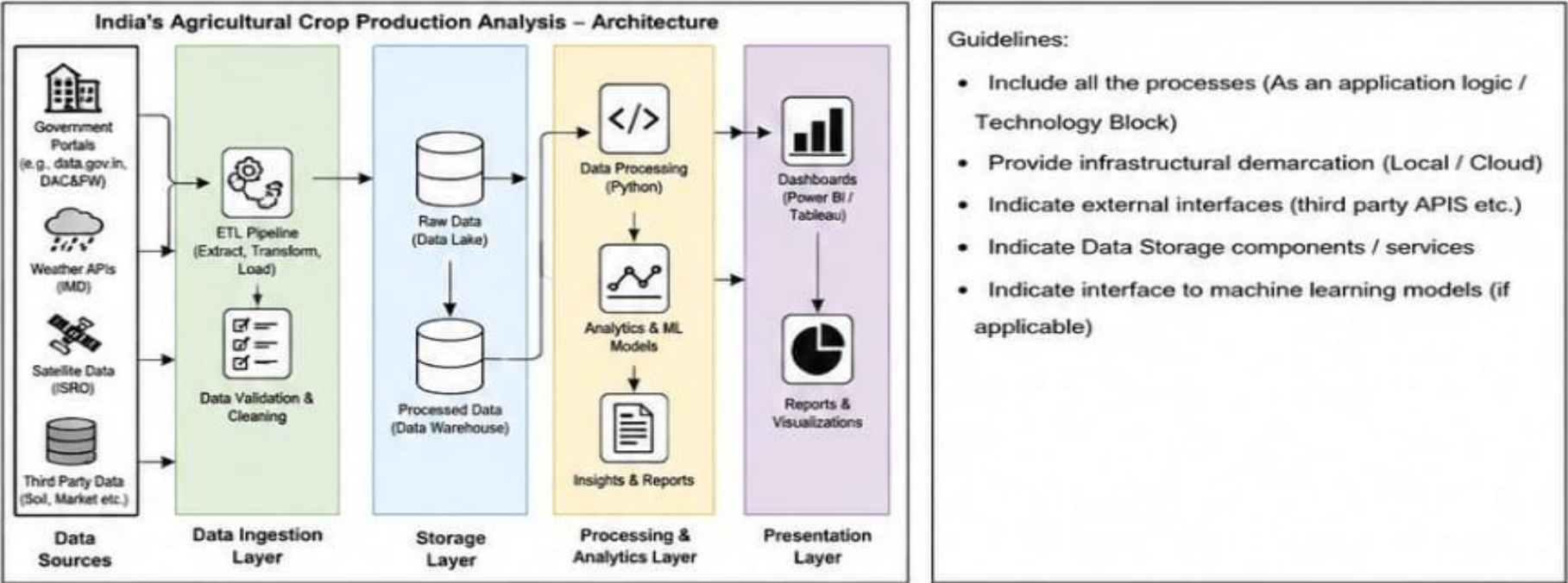


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interactive dashboards and visualizations for crop production insights	HTML, CSS, JavaScript, Power BI / Tableau
2	Application Logic-1	Core business logic for data processing, analysis and reporting	Python (Django/Flask)
3	Application Logic-2	Data aggregation, calculation of metrics (area, production, yield)	Python (Pandas, NumPy)
4	Application Logic-3	ML models for yield prediction and insight generation	Python (Scikit-learn, XGBoost)
5	Database	Structured data storage for crop statistics, reference data	PostgreSQL / MySQL
6	Cloud Database	Scalable storage for large datasets	Amazon S3 / Google Cloud Storage
7	File Storage	Storage for raw datasets, reports, and documents	Amazon S3 / Google Cloud Storage
8	External API-1	Weather data integration	IMD Weather API
9	External API-2	Market price data integration	Agmarknet / eNAM API
10	Machine Learning Model	Crop yield prediction, trend analysis	Scikit-learn, XGBoost, Python
11.	Infrastructure (Server / Cloud)	Deployment and hosting of application and databases	AWS / Google Cloud (EC2, S3, RDS, Cloud Functions)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Use of open-source frameworks for development and analytics	Django, Flask, Pandas, Scikit-learn, Bootstrap
2	Security Implementations	User authentication, role-based access control, data encryption	JWT, HTTPS, AES Encryption, IAM

S.No	Characteristics	Description	Technology
3	Scalable Architecture	Scalable architecture to handle large volume of agricultural data	Microservices, Docker, Kubernetes
4	Availability	High availability of application and data with backups	AWS RDS Multi-AZ, S3 Redundancy
5	Performance	Optimized queries, caching, and efficient data processing	Redis Cache, Query Optimization, Pandas, Dask